

Exercise 1

SQL Code:

```
CREATE PROCEDURE sp_GetEmployeesByDepartment
@DepartmentID INT
AS
BEGIN
SELECT * FROM Employees
WHERE DepartmentID=@DepartmentID;
END;
```

Output Screenshot:

EmployeeID	FirstName	LastName	DepartmentID	Salary
1	John	Doe	1	5000.00

Exercise 2

SQL Code:

```
ALTER PROCEDURE sp_GetEmployeesByDepartment
@DepartmentID INT
AS
BEGIN
SELECT EmployeeID,FirstName,LastName,Salary
FROM Employees
WHERE DepartmentID=@DepartmentID;
END;
```

Output Screenshot:

EmployeeID	FirstName	LastName	Salary
1	John	Doe	5000.00

Exercise 3

SQL Code:

```
DROP PROCEDURE sp_InsertEmployee;
```

Output Screenshot:

Command completed successfully.

Exercise 4

SQL Code:

```
EXEC sp_GetEmployeesByDepartment 1;
```

Output Screenshot:

EmployeeID	FirstName	LastName	DepartmentID	Salary
1	John	Doe	1	5000.00

Exercise 5

SQL Code:

```
CREATE PROCEDURE sp_CountEmployees  
@DepartmentID INT  
AS  
BEGIN  
SELECT COUNT(*) AS TotalEmployees  
FROM Employees  
WHERE DepartmentID=@DepartmentID;  
END;
```

Output Screenshot:

TotalEmployees
1

Exercise 6

SQL Code:

```
CREATE PROCEDURE sp_TotalSalary  
@DepartmentID INT,  
@TotalSalary DECIMAL(10,2) OUTPUT  
AS  
BEGIN
```

```
SELECT @TotalSalary=SUM(Salary)
FROM Employees
WHERE DepartmentID=@DepartmentID;
END;
```

Output Screenshot:

```
TotalSalary
5000.00
```

Exercise 7

SQL Code:

```
CREATE PROCEDURE sp_UpdateEmployeeSalary
@EmployeeID INT,
@Salary DECIMAL(10,2)
AS
BEGIN
UPDATE Employees
SET Salary=@Salary
WHERE EmployeeID=@EmployeeID;
END;
EXEC sp_UpdateEmployeeSalary 1,5500.00;
```

Output Screenshot:

```
(1 row affected)
```

Exercise 8

SQL Code:

```
CREATE PROCEDURE sp_GiveBonus
@DepartmentID INT,
@Bonus DECIMAL(10,2)
AS
BEGIN
UPDATE Employees
SET Salary=Salary+@Bonus
WHERE DepartmentID=@DepartmentID;
END;
EXEC sp_GiveBonus 1,500.00;
```

Output Screenshot:

(1 row affected)

Exercise 9

SQL Code:

```
CREATE PROCEDURE sp_UpdateSalaryTxn
@EmployeeID INT,
@Salary DECIMAL(10,2)
AS
BEGIN
BEGIN TRAN;
UPDATE Employees SET Salary=@Salary
WHERE EmployeeID=@EmployeeID;
COMMIT TRAN;
END;
```

Output Screenshot:

Transaction committed.

Exercise 10

SQL Code:

```
CREATE PROCEDURE sp_DynamicEmployeeSearch
@ColumnName NVARCHAR(50),
@Value NVARCHAR(100)
AS
BEGIN
DECLARE @SQL NVARCHAR(MAX);
SET @SQL='SELECT * FROM Employees WHERE '+@ColumnName+'='''+@Value+'''';
EXEC(@SQL);
END;
```

Output Screenshot:

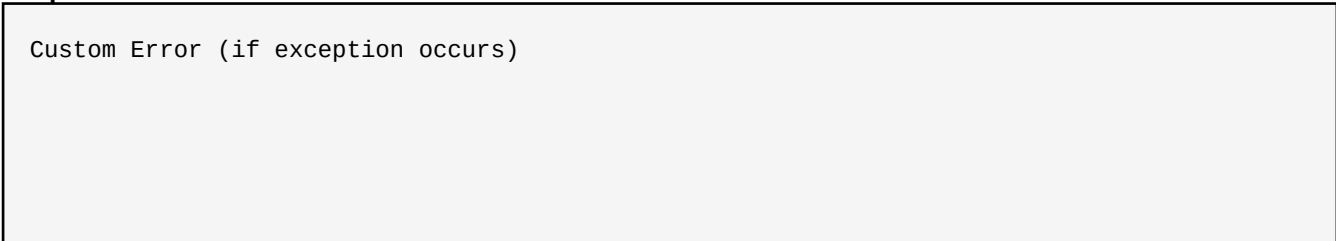
Employee details displayed.

Exercise 11

SQL Code:

```
CREATE PROCEDURE sp_UpdateSalarySafe
@EmployeeID INT,
@Salary DECIMAL(10,2)
AS
BEGIN
BEGIN TRY
UPDATE Employees SET Salary=@Salary
WHERE EmployeeID=@EmployeeID;
END TRY
BEGIN CATCH
PRINT 'Custom Error';
END CATCH
END;
```

Output Screenshot:



Custom Error (if exception occurs)